



DG INTERNS HUB

Cybersecurity Internship

WEEK 1 ASSIGNMENT

NETWORKING FUNDAMENTALS

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Internship	Cybersecurity Internship
Week	Week 1
Assignment:	Networking Fundamentals

Topics Covered: OSI Model, TCP/IP Model, Network Protocols, IP Addressing & Subnetting, Network Commands, and Cisco Packet Tracer Simulation.

1. Introduction & Objectives

Computer networking is an important part of cybersecurity. A network allows computers and other devices to communicate and share data with each other. To understand cybersecurity, it is important to first understand how data travels through a network.

In this Week 1 assignment, I learned the basic concepts of computer networking. I studied the OSI Model, TCP/IP Model, common network protocols, IP addressing, and basic subnetting.

I also performed practical tasks using Windows Command Prompt and Cisco Packet Tracer. These activities helped me understand how network devices communicate with each other.

1.1 Objectives

- Understand the 7 layers of the OSI Model.
- Learn the 4 layers of the TCP/IP Model.
- Understand HTTP, HTTPS, FTP, DNS and DHCP.
- Understand IP addresses, public/private IP and basic subnetting.
- Practice ping, ipconfig and tracert commands.
- Build a PC - Switch - Router network in Cisco Packet Tracer.
- Test connectivity using ping.

1.2 Why Networking Matters in Cybersecurity

Networking knowledge helps cybersecurity professionals understand how devices communicate and how data moves from one system to another. It is also useful for identifying network problems and understanding suspicious network activity. Learning networking fundamentals is therefore an important first step in my cybersecurity learning journey.

2. Task 1 - OSI Model

The **OSI (Open Systems Interconnection) Model** explains how data travels from one device to another over a network. It divides network communication into seven layers, and each layer performs a specific function.

Layer	Name	Function	Real-life example
7	Application	Provides network services to applications used by the user.	Opening a website in a web browser.
6	Presentation	Handles data formatting, encryption and compression.	Encrypting data while using a secure website.
5	Session	Establishes, manages and ends communication sessions.	Starting and ending a communication session between two systems.
4	Transport	Provides end-to-end delivery and divides data into smaller segments.	TCP helping data reach its destination reliably.
3	Network	Handles IP addressing and routing of packets.	A router selecting a path to send packets.
2	Data Link	Handles frames and MAC addresses on a local network.	A switch forwarding data to the correct device.
1	Physical	Transmits raw bits through physical or wireless media.	An Ethernet cable carrying network signals.

Simple Example

When I open a website, the request passes through different networking layers. The Application Layer provides the network service, the Transport Layer helps deliver the data, the Network Layer uses IP addresses for routing, and the lower layers help transmit the data through the network.

What I learned: The OSI Model makes network communication easier to understand by separating it into different layers. It also helps identify where a networking problem may occur.

3. Task 2 - TCP/IP Model

The **TCP/IP (Transmission Control Protocol/Internet Protocol) Model** is used for communication over the Internet and other networks. It has four layers.

Layer	Name	Function	Example
4	Application	Provides network services to applications.	HTTP/HTTPS when opening a website.
3	Transport	Provides communication using TCP or UDP.	TCP can provide reliable delivery of web data.
2	Internet	Handles IP addressing and routing.	A router forwards packets using IP addresses.
1	Network Access	Handles local-network and physical transmission.	Ethernet or Wi-Fi connection.

3.1 OSI Model vs TCP/IP Model

The OSI Model has seven layers, while the TCP/IP Model has four layers. In TCP/IP, some OSI layers are combined.

OSI Model	TCP/IP Model
Application	Application
Presentation	Application
Session	Application
Transport	Transport
Network	Internet
Data Link	Network Access
Physical	Network Access

What I learned: Both models explain network communication. OSI provides a seven-layer reference structure, while TCP/IP uses four practical layers.

4. Task 3 - Common Network Protocols

Network protocols are rules that allow devices to communicate and exchange data over a network. In this task, I studied HTTP/HTTPS, FTP, DNS and DHCP.

Protocol	What it does	Common port(s)	Where used
HTTP / HTTPS	Transfers web data. HTTPS protects communication using encryption.	80 / 443	Websites and web applications.
FTP	Transfers files between computers over a network.	20 / 21	File upload/download and server file transfer.
DNS	Converts domain names into IP addresses.	53	Finding the IP address for names such as google.com.
DHCP	Automatically provides network settings such as an IP address.	UDP 67 / 68	Home, office and other networks with automatic addressing.

Examples

HTTP/HTTPS: When I open a secure website, HTTPS helps protect the data exchanged between my browser and the website.

FTP: FTP can be used to upload files from a computer to a server.

DNS: When I enter google.com in a browser, DNS helps find the IP address associated with that domain.

DHCP: When a laptop joins a network, DHCP can automatically provide its network configuration.

What I learned: HTTP/HTTPS are used for web communication, FTP transfers files, DNS converts domain names into IP addresses, and DHCP automatically provides network settings.

5. Task 4 - IP Addressing & Basic Subnetting

An **IP (Internet Protocol) Address** is a numerical address used to identify a device on a network. An IPv4 address contains four numbers separated by dots, for example **192.168.1.10**. IPv4 addresses are 32 bits long.

5.1 Public and Private IP Addresses

Public IP	Private IP
Used on the public Internet	Used inside a local network
Globally unique	Used within private networks
Publicly routable	Not directly routable on the public Internet
Usually provided for Internet connectivity	Can be assigned within a private network

Common Private IPv4 Ranges

Private IPv4 range	CIDR
10.0.0.0 - 10.255.255.255	/8
172.16.0.0 - 172.31.255.255	/12
192.168.0.0 - 192.168.255.255	/16

5.2 Basic Subnetting

Subnetting means dividing a network into smaller subnetworks called subnets. In my Packet Tracer practical, I used the subnet mask **255.255.255.0**, which can also be written as **/24**. A /24 network has 256 total addresses and 254 usable host addresses.

What I learned: I understood the difference between public and private IP addresses and learned how a subnet mask separates the network and host portions of an IPv4 address.

6. Task 5 - Practical Network Commands

I performed the following commands on my own Windows laptop and captured the results as practical evidence.

6.1 ipconfig

The **ipconfig** command displays the network configuration of a Windows computer. From my output, I observed IPv4 address **10.146.5.37**, subnet mask **255.255.255.0**, and default gateway **10.146.5.241**.

```
Microsoft Windows [version 10.0.20200.9100]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Akash>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : 2409:40d1:144c:13:a4f5:8f61:
    Temporary IPv6 Address. . . . . : 2409:40d1:144c:13:4dc:eab5:b
    Link-local IPv6 Address . . . . . : fe80::cbf:463:1514:b35c%17
    IPv4 Address. . . . . : 10.146.5.37
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::2884:5fff:fe8b:87a3%17
                                10.146.5.241
```

Figure 1. My actual ipconfig output from Windows Command Prompt.

6.2 ping google.com

I used **ping google.com** to test network connectivity. Four packets were sent and all four were received, giving **0% packet loss**. The average round-trip time shown in my test was **43 ms**.

```
C:\Users\Akash>ping google.com

Pinging google.com [2404:6800:4013:804::8b] with 32 bytes of data:
Reply from 2404:6800:4013:804::8b: time=36ms
Reply from 2404:6800:4013:804::8b: time=30ms
Reply from 2404:6800:4013:804::8b: time=59ms
Reply from 2404:6800:4013:804::8b: time=47ms

Ping statistics for 2404:6800:4013:804::8b:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 30ms, Maximum = 59ms, Average = 43ms
```

Figure 2. My actual ping test to google.com.

6.3 tracert google.com

I used **tracert google.com** to view the path taken by packets toward Google. Some intermediate hops displayed **Request timed out**, but the trace still reached the destination and completed successfully.

```
C:\Users\Akash>tracert google.com

Tracing route to google.com [142.250.29.102]
over a maximum of 30 hops:

  0  73 ms  7 ms  5 ms  dsldevice.lan [192.168.1.254]
  1  70 ms  5 ms  5 ms  103.164.199.6
  2  70 ms  19 ms  22 ms  103.164.199.5
  3  *      *      *      Request timed out.
  4  18 ms  109 ms  85 ms  72.14.223.14
  5  43 ms  82 ms  76 ms  142.251.66.169
  6  204 ms  20 ms  19 ms  192.178.83.224
  7  17 ms  140 ms  17 ms  142.251.234.11
  8  17 ms  80 ms  42 ms  142.251.194.216
  9  *      *      *      Request timed out.
 10  *      *      *      Request timed out.
 11  *      *      *      Request timed out.
 12  *      *      *      Request timed out.
 13  *      *      *      Request timed out.
 14  *      *      *      Request timed out.
 15  *      *      *      Request timed out.
 16  91 ms  84 ms  18 ms  nw-in-f102.1e100.net [142.250.29.102]

Trace complete.
```

Figure 3. My actual traceroute to google.com.

What I learned: These commands helped me check my computer's IP configuration, test connectivity, and understand that packets can travel through multiple hops before reaching a destination.

7. Task 6 - Cisco Packet Tracer Network Simulation

I built a simple network in Cisco Packet Tracer using a PC, a Cisco 2960 switch and a Cisco 1941 router. I connected the devices using Copper Straight-Through cables.

7.1 Network Topology

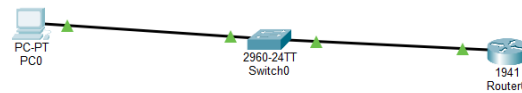


Figure 4. My Packet Tracer topology: PC0 - Switch0 - Router0. Green indicators show active links.

7.2 PC0 IP Configuration

I configured PC0 with IP address **192.168.1.10**, subnet mask **255.255.255.0**, and default gateway **192.168.1.1**.

The screenshot shows the configuration window for PC0. The 'Config' tab is selected. Under 'IP Configuration', the 'Interface' is 'FastEthernet0'. The 'Static' radio button is selected. The 'IPv4 Address' is '192.168.1.10', 'Subnet Mask' is '255.255.255.0', 'Default Gateway' is '192.168.1.1', and 'DNS Server' is '0.0.0.0'. Under 'IPv6 Configuration', the 'Static' radio button is selected. The 'IPv6 Address' is empty, 'Link Local Address' is 'FE80::209:7CFF:FE26:3817', 'Default Gateway' is empty, and 'DNS Server' is empty. Under '802.1X', the 'Use 802.1X Security' checkbox is unchecked, 'Authentication' is 'MD5', 'Username' is empty, and 'Password' is empty.

Figure 5. My actual PC0 IPv4 configuration in Cisco Packet Tracer.

7.3 Router Configuration & Connectivity Test

I configured Router0 on the connected GigabitEthernet interface with IP address **192.168.1.1** and subnet mask **255.255.255.0**, and enabled the port. After the link became active, I tested connectivity from PC0.

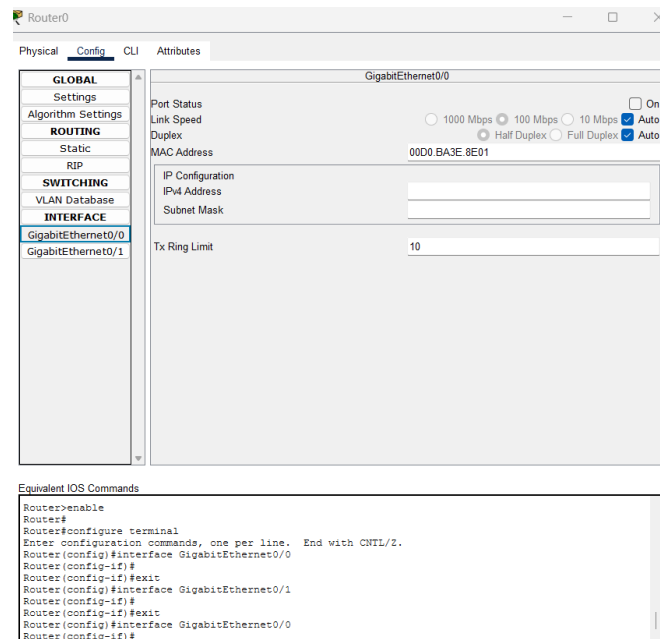


Figure 6. Router0 interface configuration screen used during my practical.

7.4 Ping Verification

From the Packet Tracer PC0 Command Prompt, I ran **ping 192.168.1.1**. All four packets were received with **0% packet loss**. This confirmed successful communication between PC0 and Router0.

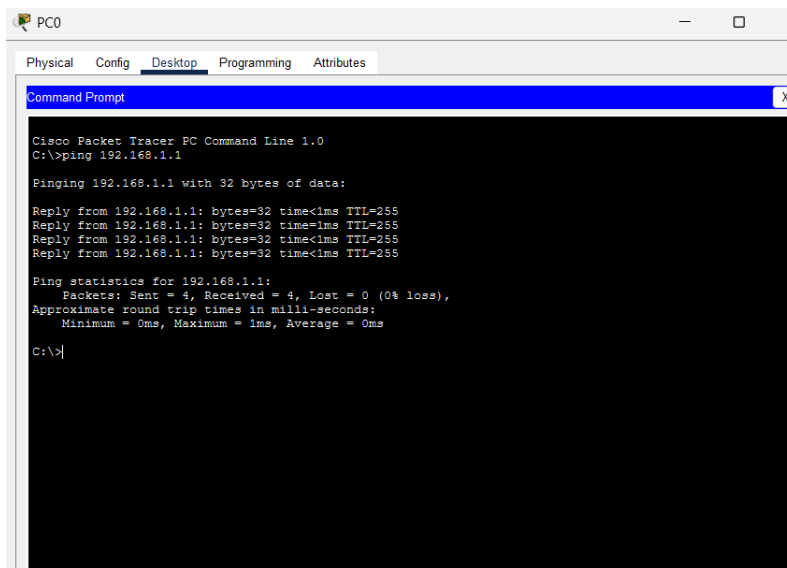


Figure 7. Successful ping from PC0 to Router0 in Cisco Packet Tracer.

8. Key Learnings & Conclusion

During Week 1, I learned the basic concepts that form the foundation of computer networking. I understood how the OSI and TCP/IP models organize network communication, and I learned the purpose of common protocols such as HTTP/HTTPS, FTP, DNS and DHCP.

I also learned the difference between public and private IP addresses and understood the basic idea of subnetting. The command-line practical gave me experience with ipconfig, ping and tracert on my own laptop.

The Cisco Packet Tracer activity was especially useful because I created a PC - Switch - Router network myself, configured IPv4 settings, enabled the router interface and verified connectivity using ping.

8.1 Main Takeaways

- Networking models help organize and understand communication between devices.
- Different protocols perform different tasks on a network.
- IP addresses identify devices, while subnet masks define network boundaries.
- ping, ipconfig and tracert are useful basic diagnostic commands.
- A switch connects devices on a local network, while a router works with IP networks.
- Correct addressing and active interfaces are necessary for successful communication.

8.2 Conclusion

Completing this Week 1 assignment improved my understanding of networking fundamentals and gave me hands-on experience with real network commands and Cisco Packet Tracer. These concepts will be useful as I continue learning cybersecurity.

Practical Evidence: All screenshots included in the practical sections were captured from the work I performed during this assignment.